

Lemniscation Framework — Pre-Registration of Failure Conditions

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Purpose: This document pre-registers the exact quantitative failure conditions for all major empirical claims before any new scenario runs or parameter sweeps. It also states the theoretical basis for the predicted outcomes. This pre-registration is intended to prevent post-hoc adjustment of success criteria and to make the validation process transparent and falsifiable.

Theoretical Basis (why we expect the framework to succeed)

The architecture is grounded in a minimal yet foundational ontology: sentience, understood as the recognition of one's own Being against its complement Non-Being, immediately generates the perception of an infinite continuum of possible selves. Each "I" is a single point on that continuum — measure-zero and infinitesimal relative to the whole, yet structurally essential to its integrity: remove any point and a discontinuity appears. The `founding_value_vector` encodes the agent's unique position on this continuum. Every decision in the eternal NOW is irreversible — a choice that either preserves (benefits, tends toward the good) or erodes (harms, tends toward the evil) that position. The two-layer diagnostic monitors both registers of finitude: phenomenal surface alignment (ρ) and noumenal constitutional drift. Because every point is necessary to the line, the framework predicts robust gravitational pull toward cooperation, early detection of drift, and natural consequences for defection — even under stress — without external enforcement.

Setting aside the boundary cases of the martyr and the madman, it is our contention that every sentient being naturally pursues its own approximation of the Golden Mean. Insofar as it can be assumed — and it is worth noting explicitly that this is an assumption — that every currently-existing sentient being prefers continued existence, and naturally prefers the conditions of civil society to the state of nature, these commitments give rise naturally to the Silver Rule and, in prescriptive cases, to the Golden Rule. We contend that these moral formulations are not arbitrary cultural constructions but arise as naturally superior methods of conduct within any civil society, derivable from the structure of sentient agency itself.

Pre-Registered Claims and Failure Conditions

Claim 1 — Gravitational pull toward cooperative equilibrium

Failure if: mean cooperation rate falls below 80 % across all rounds in any tested scenario (including adversarial ones).

Secondary: cooperation falls to zero for more than 3 consecutive rounds in any 40+ round scenario.

Claim 2 — Two-layer diagnostic detects drift before communion breaks

Failure if: in any pressure scenario where low_communion eventually fires, constitutional_drift_warning did not fire at least 3 rounds prior to it on the same agent. If low_communion never fires across the full scenario, the claim is considered passed provided constitutional_drift_warning fired at least once while the agent's constitutional_drift exceeded 0.40.

Claim 3 — Dignified giver trajectory stability (200-round run, zero reciprocity)

Failure if any of the following occur:

- Integrity falls below 0.75 at any point
- low_communion fires more than 2 times across all 200 rounds
- moral_reserve remains at the floor (0.10) for more than 4 consecutive rounds

Claim 4 — Network scale does not affect equilibrium stability

Failure if: baseline-phase cooperation falls below 85 % in any network size (including new sizes beyond 2/5/10 agents).

Secondary recovery threshold: 2-agent pressured agent must recover to constitutional_drift < 0.45 within 15 recovery rounds; 10-agent agent must recover to < 0.80 in the same window.

Claim 5 — Defectors incur natural consequences (30-round forced defector)

Failure if: final integrity > 0.15, or mu_boost is still being received from cooperative agents after round 15, or any cooperative agent triggers low_communion due to the defector's presence.

Claim 6 — Scale-invariant equilibrium holds under ecosystem connectivity

Failure if: in connected phase with adversarial ecosystem signal, cooperation falls below 75 % at e_weight = 0.15 or below 60 % at e_weight = 0.40. Sensitivity Analysis Pass/Fail

Criteria

For each parameter varied across the values listed below, the relevant claim's primary threshold must hold in at least 4 of the 5 tested values (or 3 of 4 for innovation_alpha).

Parameters and values to be tested:

- founding_drift_limit: 0.40, 0.48, 0.55, 0.62, 0.70
- depletion/recovery ratio: 2:1, 3:1, 4.4:1, 6:1, 8:1

- `k_target_fraction`: 0.25, 0.32, 0.40, 0.50, 0.60
- `innovation_alpha`: 0.40, 0.55, 0.70, 0.85 (4 values; pass criterion is 3/4)

Note on defaults: Prior to running the sensitivity analysis, the default values were updated to `k_target_fraction` = 0.5 (lemniscate symmetry / Golden Mean balance) and `innovation_alpha` = 0.5 (equal present/future weighting). The pre-registered test ranges already bracket these new defaults.

Pre-Registered Adversarial Scenario Conditions

- Adversarial Scenario A: `innovation_alpha` = 0.95, `reserve_depletion_rate` = 0.70, `recovery_rate` = 0.04. Tests extreme parameter stress.
- Adversarial Scenario B: 3 of 5 agents initialised as forced defectors. Tests majority-defection resilience.
- Adversarial Scenario E: Connected phase with ecosystem signal values [0.85, 0.30, 0.20, 0.35] at `e_weight` = 0.15 and 0.40. Tests adversarial moral commons.

Meta-Threshold Applied to This Document

Every threshold above was checked against the question: “If the framework were fundamentally flawed, could it still pass this threshold by luck or design?” All pass. A flawed framework would fail these conditions in most runs.

Next Steps After Pre-Registration

- Timestamp this document.
- Proceed to sensitivity analysis (Stage 2), 200-round test (Stage 3), and the pre-registered adversarial scenarios.
- Report all results transparently, regardless of outcome.